A.3 VISUALIZATIONS OF INTERNAL ACTIVATIONS

We visualize the internal activations for a string template "x1 + x2 =" on the second numeric token (x2). Values for x2 are selected as a range 0-999 (all values present by the vocab), and values for x1 are sampled randomly from the same range. Then, we project the activations of the model's middle layer to 64 dimensions with PCA and compute the Fourier transform.

We visualize the first 16 PCA dimensions in Figure 14 and the maximal magnitudes of the frequencies in the Fourier transform in Figure 15. We visualize across model sizes and families.

B NATURAL LANGUAGE DATASET DESCRIPTION

Table 1: Dataset specifications for numerical embedding analysis across natural language contexts.

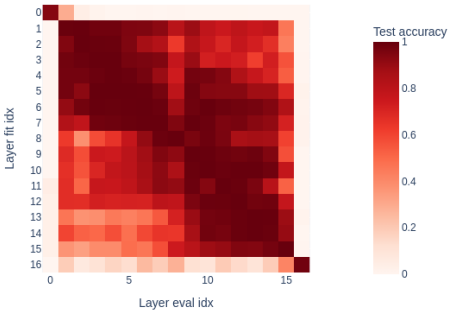
Domain	Dataset	Source	Numerical Context	Size
Culinary	Recipe NLG Lite	`m3hrdadfi/recipe_nlg_lite`	Quantities, measurements	6118
	FoodRecipe-ImageCaptioning	samsatp/FoodRecipe-ImageCaptioning	Ingredient amounts	719
Temporal	TimeLineExtraction	irlabamsterdam/TimeLineExtraction...CASE	Legal document dates	50
Arithmetic	MetaMathQA	`meta-math/MetaMathQA`	Mathematical reasoning	395K
	DROP	`ucinlp/drop`	Discrete reasoning	77.4K
	AQuA-RAT	`deepmind/aqua_rat`	Algebraic word problems	97.4K
Medical	ICD-10 Codes	`atta00/icd10-codes`	Diagnostic codes	25.7K
	ICD-10-CM	`Gokul-waterlabs/ICD-10-CM`	Medical classifications	74K

B.1 QUALITATIVE ASSESSMENT OF OUTPUT EMBEDDINGS

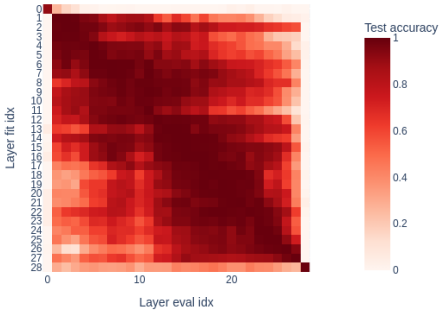
We visualize output representations of Llama 3 1B on addition and multiplication operations. For each expected result value y in the range 0-1000, we sample a random pair (x1, x2) from the same range, such that x1+x2=y (or x1*x2=y, respectively). We then predict the next token for prompts "x1 plus x2 is " and "x1 multiplied by x2 is " and collect the final output representations of the model before decoding. We then reduce the representations with PCA to 16 dimensions and visualize the result in Figure 16.

	Add	Sub	Mul	Div
Accuracy	100%	99.8%	90.4%	5.9%
P(Extracted | Incorrect)	-	56.8%	26.3%	94.4%
P(Not extracted | Correct)	1.4%	0.03%	26.1%	5.9%

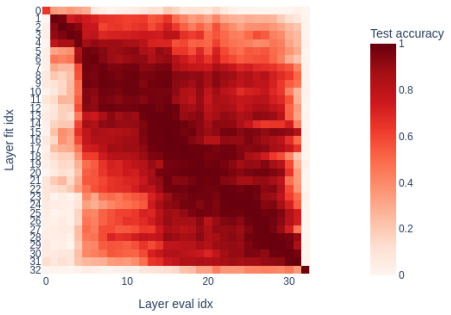
Table 2: **Probe accuracy in extracting the predicted result:** Ratio of cases where the sin probe of some layer (top) retrieves a correct result when the model's prediction is *incorrect*, and (bottom) can *not* retrieve a correct result when the model's prediction is *correct*.



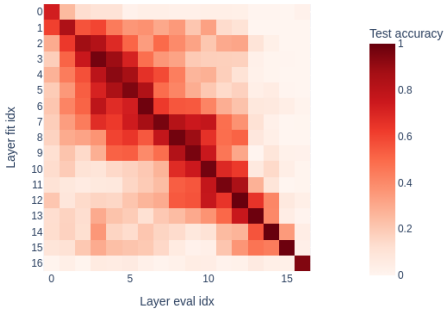
(a) Llama 3 1B



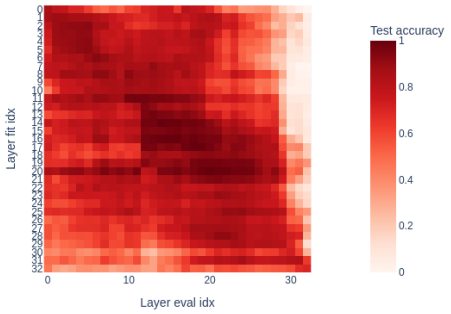
(b) Llama 3 3B



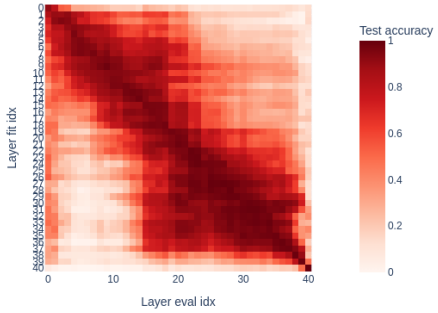
(c) Llama 3 8B



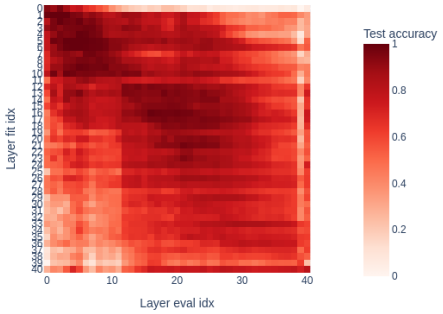
(d) Olmo 2 1B



(e) Olmo 2 7B

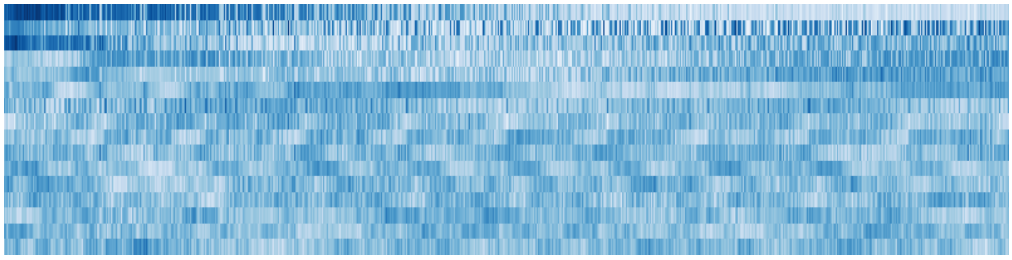


(f) Olmo 2 13B

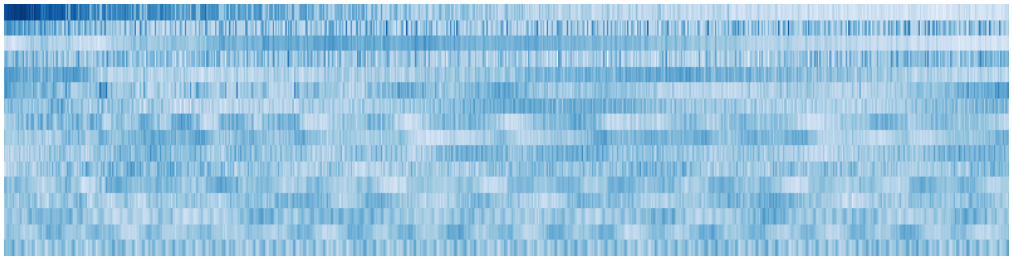


(g) Phi 4 15B

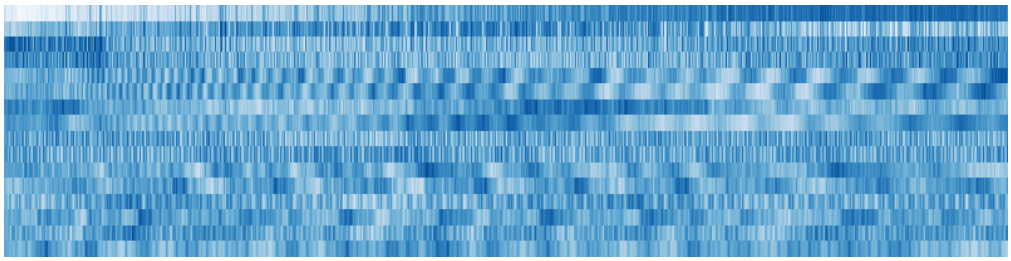
Figure 13: Probe fitted on one layer evaluated on all layers. Olmo 2 1B shows the weakest cross-layer generalization among all models. Llama models display a strong separation between input/output embedding representations and the hidden representations.



(a) Llama 3 1B (layer 8/17)



(b) Olmo 2 7B (layer 16/31)



(c) Phi 4 15B (layer 20/41)

Figure 14: PCA of models' internal representations